

Putting Quantum AI Hype to the Test in Cislunar Space: A Governed Public-Data Benchmark for Human-Centered Trajectory-Correction Planning

Taechasith Kangkhuntod

University of the Thai Chamber of Commerce, Bangkok, Thailand

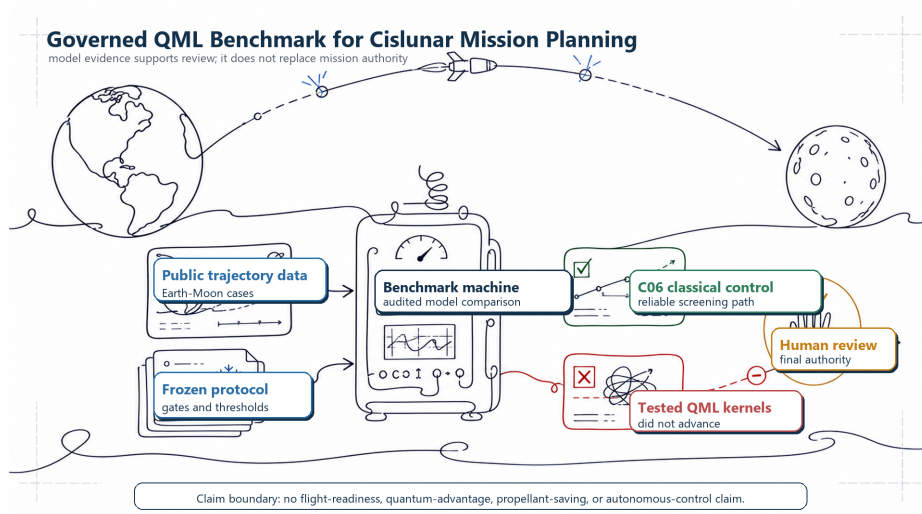


Fig. 1. Labeled one-line pen illustration of the study workflow: public trajectory data and a frozen protocol enter an audited benchmark machine, model evidence separates into a reliable C06 classical-control path and a non-advancing QML-kernel path, and final authority remains with human mission review.

Abstract. This study tests whether quantum machine learning (QML) kernels improve a public-data benchmark for crewed Earth–Moon trajectory correction. The target is robust total correction Δv : modeled velocity change after accounting for uncertainty. Related mission cases shared a data fold; preprocessing used only training rows; thresholds were fixed before outcomes; and calibration and final-test rows remained locked. The simulator chain passed 67 eligible checks; one event lacked eligible evidence and was classified neither pass nor failure. On development data, the primary fidelity kernel (Q01) produced mean normalized root-mean-square error (NRMSE) 0.646614, while the physics-residual classical control (C06) produced 0.008739. NRMSE divides root-mean-square error by the target standard deviation; lower is better. The projected-kernel follow-up (Q01b) produced 0.661207 versus 0.006833

for C06. The feasibility-only quantum kernel (FQK) retained 10.9% of actually feasible cases at the fixed 0.5 threshold. Sixteen bounded successors failed predeclared advancement criteria; the strongest apparent gain did not pass its matched classical-control threshold. This does not show that QML is universally ineffective; it shows that the tested representations did not justify quantum complexity for this frozen task. The societal contribution is a transparent safeguard against unsupported AI claims in high-consequence decisions: preserving provenance, reporting negative outcomes, and separating crew requirements and human authorization from model scores. This study makes no claim of quantum advantage, propellant savings, flight readiness, hardware performance, or a new QML invention.

Keywords: Quantum machine learning · human-centered AI · decision support · cislunar trajectory design · reproducible benchmarking · negative results

1 Introduction

Cislunar space is the region in and around the Earth–Moon system. A trajectory-correction maneuver, or correction burn, deliberately changes a spacecraft’s velocity to keep its path within mission constraints. The required change is summarized by delta-v (measured in m/s). Delta-v is not propellant mass, although it is a standard measure of maneuver effort and is related to fuel use through the vehicle and propulsion model.

For a crew-carrying mission, correction planning is a coupled decision problem: burn timing and delta-v interact with navigation uncertainty, maneuver execution, crew timelines, entry conditions, and reserve constraints. Public Artemis trajectory-design research therefore treats robust correction cost, meaning correction delta-v evaluated with modeled uncertainty, together with mission constraints rather than as an unconstrained curve-fitting task [22]. A learning *surrogate*, a fast predictive model that approximates a more expensive simulation, could help screen candidate scenarios. That possibility is a motivation for testing, not evidence that a particular machine-learning model is useful or safe.

The paper adopts a human-centered decision-support framing. A model score is not the decision: a responsible reviewer needs to know which source data, physical assumptions, uncertainty boundary, and failure cases produced the score before deciding whether a model warrants further study. Crew-related requirements therefore remain separate from predictive performance, consistent with the intended-use and human-system boundaries in the applicable U.S. National Aeronautics and Space Administration (NASA) standards [8,9]. This work does not conduct a user study or measure usability, trust, workload, or operator performance. From a human–computer interaction (HCI) perspective, its contribution is the evidence design around a possible future decision-support tool.

Quantum machine learning (QML) uses quantum-state representations or quantum circuits inside a learning method. In a quantum-kernel model, a circuit

maps each classical input to a quantum state and the overlap between two states is used as a similarity score. This offers feature maps different from common classical models [16,5], but difference alone does not imply usefulness. Encoding data introduces loading choices, similarity geometry, and resource costs. A circuit may be difficult to simulate in general while still being approximated well by a classical learner on the tested data [6,21]. Conversely, a favorable result can arise from a weakly tuned comparator, related mission cases leaking across a data split, thresholds chosen after seeing outcomes, or omitted failures. These are benchmarking problems before they are quantum-computing problems.

This paper documents a complete public-data experimental program for trajectory-correction prediction and feasibility screening. The purpose was not to assert a priori that QML would win. Its purpose was to establish a scientifically usable decision process: define a bounded task, verify the simulator against eligible public endpoints, freeze the development protocol, compare QML with strong controls, record negative outcomes, and prohibit claims that the evidence cannot support. This stance is consistent with model and simulation credibility practice [8] and with recent calls for fair QML comparisons [1,15].

The central research questions were deliberately narrow:

1. Can the primary fidelity-style quantum kernel (repository identifier Q01) improve correction-cost prediction over physics-derived and classical controls under the frozen development protocol?
2. If Q01 fails, can the projected follow-up Q01b improve the same task without changing data access or data splits?
3. Can the feasibility-only quantum kernel (FQK) meet its fixed screening criteria for identifying independently propagated feasible cases?
4. Which conclusions remain valid when calibration, final-test, mission-loop, and hardware evidence stay locked?

The answer to the first three questions is negative for the tested models and settings. The fourth question is the paper’s main methodological contribution. The work establishes a transparent evidence hierarchy that distinguishes simulator credibility, development-only model evidence, future-design lessons, and prohibited operational claims. Negative results are not treated as failed record keeping; they are the result when a hypothesis does not meet its predeclared standard.

The paper makes six contributions:

1. A source-bound cislunar benchmark protocol, meaning that inputs are tied to identified public files and recorded source versions, with grouped development folds, training-fold-only transformations, and retained numerical feasibility outcomes.
2. A simulator credibility chain that reports its eligible scope and its source-ineligible exception rather than converting missing evidence into a pass or failure.
3. A preregistered QML comparison against physics-residual, random-feature, compressed-input, and classical kernel controls.

4. A closed successor ledger containing exploratory and prospectively bounded negative candidates, including classical controls that replace the quantum component while keeping the surrounding construction matched.
5. A self-contained evidence and figure package that supports review of the negative conclusion without widening the study’s scientific claim.
6. A human-centered claim boundary that keeps model evidence, crew-related requirements, and future human authorization distinct.

2 Scientific Scope and Claim Boundary

The study is an offline, public-data, development-only benchmark. It is not a mission design authority, a flight certification exercise, or a performance assessment of a quantum processor. A simulation result may be useful for evaluating a model family, but it cannot by itself establish the performance of a mission-owned trajectory system or a hardware quantum implementation. The distinction is essential for a human-rated application, where human-system requirements are separate from a surrogate’s regression score [9].

2.1 Reader guide to labels and data stages

Short identifiers are retained because they link the paper to immutable repository records. They are audit keys, not scientific concepts that a reader is expected to know in advance:

Gate. A predefined authorization checkpoint. Gate 3 asks whether the simulator is credible within its public-data scope; Gate 5 evaluates models on development data; Gate 6 would be a separately authorized mission-level experiment and was never opened.

Decision and protocol IDs. A label beginning with D identifies a governance decision or bounded campaign, while P identifies a prospective protocol. For example, D046 is the decision record for one successor test. These labels are not data samples or performance grades.

Model IDs. C labels a classical candidate, Q labels a QML candidate, and A labels an analysis control. Thus C06 is the physics-residual classical control and Q01 is the primary quantum-kernel candidate.

Figure IDs. RFIG identifies the repository’s provenance record for a figure. It allows the plotted data and generating script to be found; it has no statistical meaning.

Development-only. Development rows may be used to choose and compare models. Calibration rows would adjust a frozen model’s output, and final-test rows would estimate performance only after all choices were fixed. This study read neither calibration nor final-test rows for the reported campaigns.

Fold, seed, and rung. A fold is one group-preserving train/validation partition. A seed is a fixed random-number starting point used to repeat a model reproducibly. A rung is one sample-size stage in the prespecified 128–1,024-row screening sequence.

Table 1. Claim boundary used throughout the manuscript. This table is derived from the committed claim-boundary record and does not add a new experimental result.

Evidence category	Permitted interpretation
Gate 3: simulator credibility	Simulator agreement for eligible public-data checks, not flight validation. RTC3 lacked eligible post-event evidence, so it was not tested and is neither a pass nor a failure.
Gate 5: development benchmark	Under the frozen development benchmark, the tested QML candidates did not outperform strong classical controls.
Post-Gate follow-ups	Projected-kernel protocol P001 and successor decisions D034–D049 are development-only negatives. They do not revise Gate 5 or authorize Gate 6.
Screening lesson	The recall-first classical analysis is a future protocol-design lesson, not a deployed safety filter or safety result.
Prohibited claims	No quantum advantage, propellant saving, mission-loop validity, flight readiness, NASA approval, hardware behavior, final-test result, or QML invention claim.

2.2 Human-centered decision-support stance

The proposed role of a model in this benchmark is limited to offline screening information for qualified human review. It is not an autonomous controller, an operational recommendation engine, or a replacement for mission analysis. For an HCI-oriented reading, the central design requirements are traceability, understandable scope, visible uncertainty, retained failures, and a clear fallback to physical analysis. The evidence package supports those requirements by showing the data boundary, simulator checks, controls, stopping decisions, and excluded evidence pathways together with the metrics.

No human participants, operator interfaces, observational studies, or measures of trust, workload, situation awareness, or decision quality were collected. The paper therefore makes no empirical HCI performance claim. It instead describes how a high-consequence AI research artifact can be structured so that later human evaluation is not pre-empted by an unsupported model claim.

The evidence boundary was defined before reporting. Gate 3 supports simulator credibility only for the eligible comparisons. Gate 5 supports a technical decision for a frozen development benchmark. The subsequent projected-kernel protocol (P001) and 16 bounded successor decisions (D034–D049) are development-only follow-ups; they cannot retroactively change the Gate 5 decision. Calibration and final-test rows, mission-loop scenarios, quantum-hardware or graphics-processor execution, threshold application to real operations, trained-model release, and Gate 6 remain outside the study. Table 1 summarizes the resulting language constraints.

Figure 2 records the governance state rather than a model metric. It is included because a benchmark conclusion depends on which evidence was autho-

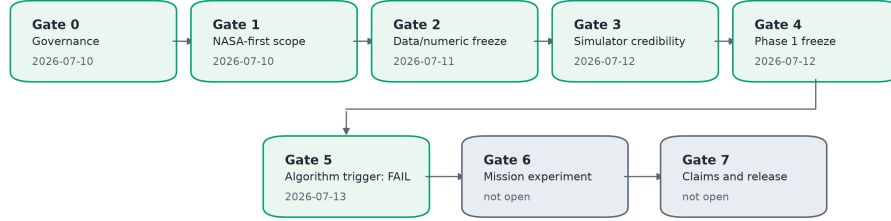
Research governance decision diagram through Gate 5

Diagram only: dates and decision states do not imply scientific success.

Fig. 2. Research governance through Gate 5 (RFIG-001). Gates 0–5 were closed by human decisions; the technical Gate 5 result is a benchmark-specific protocol FAIL, meaning its advancement hypothesis was rejected. Gate 6 remains unopened. This is decision-state evidence, not a model performance plot.

rized when. A later candidate may be scientifically informative, but it does not become preregistered evidence merely because it is reported after a negative primary result.

3 Related Work and Positioning

The target problem is positioned relative to public work on Artemis trajectory-correction design, rather than being presented as a replacement for mission-owned analysis. Woffinden et al. describe a correction-burn placement formulation that incorporates uncertainty, crew-related timing, and entry constraints [22]. Classical learning research has explored low-thrust guidance and cislunar transfer design, but those settings differ in dynamics, propulsion, objectives, and validation structure [24,20,7,19,23,25]. They are valuable precedents for surrogate modeling, low-fidelity references, cost–reachability endpoints, and retaining infeasible cases, not interchangeable baselines for a human-rated correction benchmark.

Quantum feature-space methods encode a classical input as a quantum state and compare encoded states through a quantum-derived similarity, called a kernel, [16,5]. The method can be used in a kernel learner, but its predictive value depends on whether that similarity reflects differences that matter for the target being predicted. Theoretical and numerical work has made this caveat increasingly explicit: hard-to-simulate circuits do not automatically give a learning advantage, and the data distribution can make a classical approximation competitive [6,21].

The present work is particularly informed by three benchmarking lessons. First, kernel bandwidth and encoding scale change the effective inductive bias,

meaning the patterns that a model is structurally predisposed to represent; they are not negligible implementation details [17]. Second, a quantum-kernel study should include appropriately strong classical counterparts, including random features that approximate a distance-based classical kernel where relevant [10,14,21]. Third, a comparison should report more than its best run: model choice, split structure, seed variation, capacity, and post-outcome choices all affect what a claimed gain means [1,15,12].

3.1 Literature provenance of the frozen protocol

The literature set was used to shape the protocol, not to claim that every reviewed model was executed. Data re-uploading, which repeatedly inserts input features between circuit layers, encoding-dependent generalization, and initialization studies informed the need to freeze encoding, observables, initialization, and seed handling [11,2,4]. Quantum-agent work informed the need for matched classical capacity and explicit readout choices, but it did not turn this supervised benchmark into a reinforcement-learning study [18]. Quantum annealing for trajectory transcription was retained as adjacent quantum-optimization context and deliberately excluded from the QML candidate set [3].

The accompanying literature-source audit accounts for all 23 accepted records used in the experiment’s review package and additionally identifies the random-Fourier source behind the A01 control. It distinguishes direct methodological influence from an empirical comparison in this paper. This prevents a citation from being mistaken for evidence that a reviewed external method was reproduced, validated, or outperformed here.

This paper does not make an exhaustive literature-review claim. The repository preserves a bounded extraction set of 23 accepted records used to establish the initial protocol and a separate archival discovery refresh. The refresh did not change the experiment, thresholds, figures, or outcomes. Literature is therefore used here to position the benchmark and justify controls, not to construct a retrospective theory of success.

4 Materials and Methods

4.1 Scenario construction and simulator credibility

The public pipeline generated correction scenarios at three frozen levels of physical detail. F0 uses central Earth point-mass gravity. F1 adds the Moon and Sun as third-body gravitational sources. F2 additionally includes Earth J2, the leading correction for Earth’s equatorial bulge, and tighter numerical integration settings. Across these levels the implementation retained the configured maneuver model, mass depletion, event handling, crew-related constraints, uncertainty variables, and numerical feasibility checks. The intent was to predict benchmark-defined cost or feasibility outcomes, not to control a spacecraft. Source files and scenario construction were frozen before the model campaigns; later reports could

summarize outputs but could not replace source evidence to make a comparison pass.

Simulator credibility was assessed in Gate 3 by comparing eligible Python trajectory endpoints with NASA’s General Mission Analysis Tool (GMAT) under the same accepted force-model configurations. An endpoint is the simulated position and velocity at the end of a validation window. The diagnostic chain isolated a formatting problem in the file containing the Earth-J2 coefficient and confirmed the repaired format through degree-two, order-zero gravity comparisons. In the final accepted evidence, all 67 evaluable checks passed; the repaired endpoint set also passed the unchanged position and velocity limits. This supports the implementation only within the tested public scope; it does not validate a complete mission environment.

One record, the third return-trajectory-correction event (RTC3), is explicitly excluded from the claim. Its qualified public trajectory file was created before the event and therefore cannot show the event’s reconstructed outcome under the frozen temporal-evidence rule. It is therefore classified as `not_eligible`, rather than as a successful or failed endpoint. This distinction prevents an important category error: source eligibility is not a numerical outcome. The exclusion remains visible in the evidence package rather than being silently omitted.

4.2 Frozen development data and leakage control

Let $\mathcal{D} = \{(x_i, y_i, z_i, g_i)\}_{i=1}^N$ denote the frozen development dataset, where N is the total number of development rows and i indexes one simulated candidate plan. The vector x_i contains only the allowed scenario features. The target y_i is robust total correction delta-v in m/s. The binary label z_i equals one when an independent propagation marks the plan feasible and zero otherwise. Finally, g_i identifies the related mission-design group. Whole groups, rather than individual rows, are assigned to one of five development folds. Consequently, related cases never appear on both the training and validation sides of the same comparison.

All transformations are learned within the training partition of each fold. These include filling missing values, converting categories to numbers, rescaling features and targets, and principal-component analysis (PCA), a linear compression of correlated features, where configured. Fitting any of these steps to all rows would let information from the held-out validation fold influence training. Preventing that leakage makes the comparison interpretable.

The development campaign uses nested, deterministically selected sample runs of 128, 256, 512, and 1,024 training rows. Poorly performing configurations stop at an early rung under a rule fixed before results; selected configurations are then repeated with 20 frozen random seeds. These runs are not estimates of an operational mission distribution. They are a prespecified stress test of performance across group structure, sample size, and controlled randomness. Calibration and final-test data were never read by these campaigns.

4.3 Task definitions and estimands

The primary endpoint is robust correction-cost prediction. For a fitted predictor f , normalized root-mean-square error (NRMSE) is

$$\text{NRMSE}(f) = \frac{1}{s_y} \sqrt{\frac{1}{n} \sum_{i=1}^n (f(x_i) - y_i)^2}, \quad (1)$$

where n is the number of held-out predictions, $f(x_i)$ is the predicted delta- v for row i , y_i is its simulated target, and s_y is the sample standard deviation of all frozen development targets. The numerator is the usual root-mean-square error in m/s; division by s_y makes NRMSE unitless and comparable across folds. Zero is perfect and lower is better. For example, NRMSE 0.01 means that the root-mean-square prediction error is 1% of the development target’s standard deviation. The primary report pools predictions made while each fold was held out (pooled out-of-fold NRMSE). Separate fold and seed summaries are retained so that one favorable split or random run cannot hide instability. Feasibility-constrained regret, the extra correction cost caused by selecting a worse candidate plan, is reported as an interpretation metric rather than as a replacement for the frozen primary endpoint.

The classification track estimates $p_f(x) \in [0, 1]$, the predicted probability that input x is feasible under independent propagation. Its Brier score is the mean squared probability error,

$$\text{Brier} = \frac{1}{n} \sum_{i=1}^n (p_f(x_i) - z_i)^2, \quad (2)$$

where $z_i \in \{0, 1\}$ is the observed feasibility label; lower is better and zero is perfect. Area under the receiver-operating-characteristic curve (AUROC) measures ranking across all thresholds: 0.5 indicates chance-level ranking and 1.0 indicates perfect ranking. At the fixed threshold $t = 0.5$, recall is

$$\text{Recall}_{0.5} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \quad (3)$$

Here TP counts actually feasible plans correctly marked feasible, and FN counts actually feasible plans rejected by the classifier. Thus recall is the fraction of feasible options retained. Precision is the fraction of plans marked feasible that are actually feasible; an infeasible plan incorrectly admitted is a false positive. This label direction matters: low recall reduces screening utility, whereas unsafe acceptance must be assessed through precision, false-positive behavior, and independent propagation. No one metric establishes operational safety.

The Q01 family is described generically by an encoded-state kernel

$$K_q(x, x') = |\langle 0^{\otimes q} | U_\phi(x)^\dagger U_\phi(x') | 0^{\otimes q} \rangle|^2, \quad (4)$$

where x and x' are two classical scenario vectors, $q \in \{4, 6, 8\}$ is the number of simulated qubits, $|0^{\otimes q}\rangle$ is the all-zero initial state, and $U_\phi(x)$ is the frozen

circuit that encodes x . The symbol $\dagger$ denotes the conjugate transpose, which reverses the circuit for the overlap calculation. The squared overlap K_q lies between zero and one: a larger value means that two inputs look more similar under this particular encoding. It is not a probability that either trajectory is feasible. The pairwise values form a Gram matrix, a table of every input-to-input similarity, which a kernel regressor converts into predictions. Exact features, PCA dimensions, circuit settings ϕ , bandwidths (how quickly similarity changes) and regularization values (how strongly fitting is constrained) are preserved in committed configuration records. Equation (4) defines a classically simulated research representation, not a hardware implementation or quantum speedup.

4.4 Models and matched controls

The preregistered Q01 candidate is a fidelity-style quantum-kernel regressor for correction cost. Its principal comparator is the classical physics-residual model C06. If $b_{\text{low}}(x)$ is a lower-fidelity physical estimate, C06 learns a correction $r(x)$ to form

$$\hat{y}_{\text{C06}}(x) = b_{\text{low}}(x) + r(x). \quad (5)$$

This is a stronger comparator than a generic regressor because it starts from physical structure available in the benchmark and asks machine learning to predict only the remaining error, or residual. The comparison set also contains A01 random Fourier features, a finite classical approximation to a radial-basis function (RBF) kernel; an RBF kernel treats nearby feature vectors as more similar than distant ones. Compressed classical neural-network views and other frozen configurations were also included. These controls test whether a classical representation can explain an apparent quantum-feature effect [19,23,14].

After the accepted Gate 5 result, exploratory protocol P001 defined two follow-ups without reopening the primary campaign. Q01b replaces whole-state overlap with projected features: it summarizes each simulated qubit through local measurement expectations and applies a distance-based kernel to those summaries. A deterministic Nyström approximation limits the kernel to at most 256 representative training rows, called landmarks. FQK, whose repository ID means *feasibility-only quantum kernel*, applies the same projected-kernel idea only to the independently propagated feasibility label; here FQK does not mean “fidelity quantum kernel.” Both tracks retained groupwise splits, training-fold-only pre-processing, nested rungs, matched controls, and 20 seeds. They could produce new development-only evidence, but could not revise the decision already made for Q01.

The 16 successor decisions D034–D049 tightened this discipline further. Each protocol froze its candidate, target, comparator, threshold, stopping rule, and source path before fitting. Their compact model names are immutable registry identifiers; the repository stores each complete construction. Several candidates produced a lower NRMSE than C06, but that alone was insufficient. A candidate also had to beat a matched classical alternative given the same inputs and stage structure. Such a control is often called a dequantization control because it tests

whether the apparent benefit survives when the quantum-derived component is replaced by a classical one.

4.5 Human-centered decision-support boundary

The reported model outputs are treated as evidence for a future screening tool, not as a command to change a trajectory. A human reviewer would need the model estimate, the applicable scenario group, its comparator result, its uncertainty record, and an explicit statement of whether the case is inside the authorized scope. The research package preserves those items as source records and figure provenance rather than reducing them to a single leaderboard value.

This stance also defines the fallback: when a candidate is ineligible, poorly calibrated, source-ineligible, or outside the public benchmark boundary, the model is not promoted and the physical or numerical reference remains the decision basis. This is a design boundary for later human–AI interaction; it does not evaluate an interface or the behavior of mission personnel.

4.6 Statistical decision process

The preregistered Gate 5 trigger required every eligible fold and seed to complete, comparison with the strongest classical finalist and matched dequantization controls, paired uncertainty checks, and at least one qualifying regime. A *regime* is a subgroup defined before outcomes, such as a fidelity level or uncertainty family; it is not a favorable subgroup discovered after inspection. A paired comparison subtracts model errors measured on the same folds and seeds. The paired 95% bootstrap interval repeatedly resamples those matched differences to quantify uncertainty. A sign-permutation test also asks whether the observed direction could arise if positive and negative paired differences were exchangeable under no model advantage; multiplicity control limits false discoveries across repeated comparisons. The report does not add a post-outcome seed, model, split, or threshold to repair a weak result. Learning curves and all retained seed outcomes are shown because training fit or one favorable run is not adequate evidence of generalization [13,4].

The distinction between task completion and eligibility is also retained. The primary development campaign (decision D006) completed all 871 authorized computational tasks with zero software or task failures. The variational quantum regressor Q02 and hybrid quantum-residual model Q03 were not carried to larger sample rungs because too few configurations met the frozen retention rule at the first rung. Their status is therefore `not_reached_under_frozen_eligibility`. It does not mean a failed experiment, a missing computation, or a favorable result withheld from the report.

4.7 Computational-resource and reproducibility boundary

The reference compute environment was a Windows laptop with a 13th Generation Intel Core i9-13900HX central processing unit (CPU), 32 GB random-access

memory (RAM), and an NVIDIA GeForce RTX 4060 Laptop graphics processor (GPU). A statevector stores all 2^q complex amplitudes of a q -qubit state in classical memory. The experiment bounded this memory use and ran statevector tasks one at a time; the GPU and quantum hardware were not used for the reported QML evidence. The hardware description is included so that readers understand the throughput and resource constraints under which the public benchmark was executed. It is not a prescription that future studies use identical hardware.

The source release contains configuration records, seeds, data cards, figure registry entries, machine-readable tables, and reporting scripts. Every figure used in this paper is copied with its corresponding registry entry into the paper package. These measures make the evidence auditable while preserving the lock on calibration and final-test data.

5 Results

5.1 Simulator credibility result

The accepted Gate 3 record contains 67 evaluable checks and zero failures. The result includes the Earth-only diagnostic chain that localized the original J2 discrepancy to a coefficient-file delimiter problem rather than to the integrator, time system, or initial state. Once the coefficient format was repaired, the GMAT–Python comparison passed the unchanged position and velocity thresholds across the eligible windows. The appropriate conclusion is that the public simulator implements the tested scope consistently enough to support the development benchmark. It is not a flight-readiness conclusion, and the RTC3 event remains untested because no eligible post-event public source was available. RTC3 is therefore neither a pass nor a failure.

5.2 Preregistered Gate 5 benchmark

The Q01 quantum kernel did not satisfy the development trigger. Its mean NRMSE was 0.646614. The physics-residual control C06 achieved 0.008739, making Q01’s normalized error about 74 times as large. Because lower NRMSE is better, this was not a close comparison that depended on one uncertainty test. No predeclared subgroup satisfied the advance rule. Gate 5 is therefore a valid technical **FAIL** for the specified Q01 model, task, preprocessing, controls, and development population. Here **FAIL** is a protocol decision meaning that the performance hypothesis was rejected. It is not a software crash and not a claim that every possible QML method fails.

Figure 3 shows the pooled development evidence over the authorized sample-size rungs and matched PCA dimensions. A curve shows how error changes as more training rows are used. Configurations that did not advance were stopped under the prespecified retention rule. Figure 4 reports the selected-model distribution over the 20 frozen random seeds, and Fig. 5 places the pass condition next

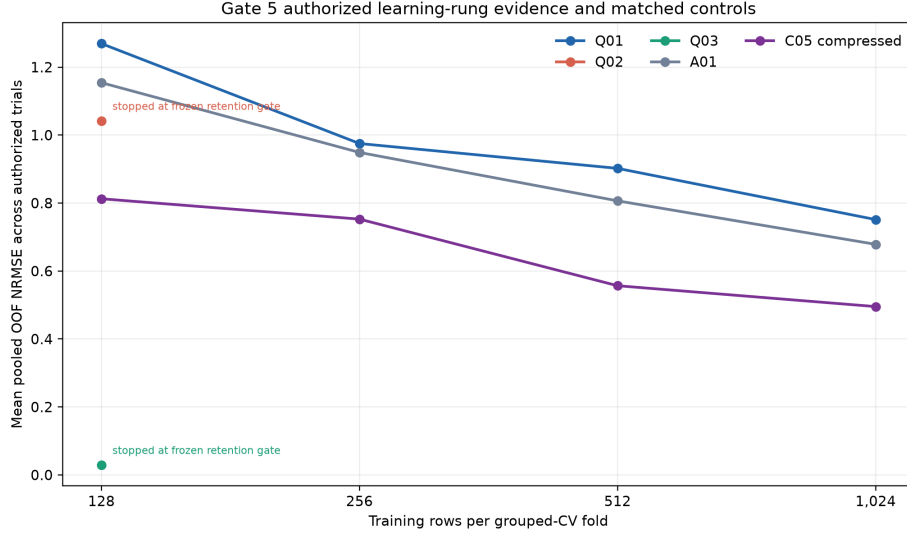


Fig. 3. Gate 5 development learning curves and matched controls (RFIG-021). The evidence covers every authorized rung and matched PCA dimension. This is development model-selection evidence only.

to the strongest classical and matched dequantization controls. The three figures should be read together: a model is not evaluated solely by a single aggregate mean, but by its retained trajectory through the precommitted decision rules.

Q02 and Q03 require a precise reading. They completed their authorized tasks, but too few configurations of the variational quantum regressor (Q02) and hybrid quantum-residual model (Q03) satisfied the frozen eligibility condition for larger rungs and seed reruns. They are recorded as “not reached under frozen eligibility” (`not_reached_under_frozen_eligibility`). Treating this state as a failed experiment would overstate the evidence; treating it as an unreported potential success would ignore the protocol. The paper reports the actual decision state.

5.3 Post-Gate 5 exploratory tracks

Q01b tested whether changing the quantum readout while keeping the same development discipline could materially alter the result. It did not. Q01b attained mean NRMSE 0.661207, compared with 0.006833 for C06, so its error was about 97 times as large. Figure 6 presents its learning curves against the source-matched controls, while Fig. 7 reports the projected-kernel geometry and dequantization diagnostics. These are useful negative data: for this projected feature map, the representation did not capture the residual structure better than the classical controls. They do not establish that projected quantum kernels cannot work on any other physical problem.

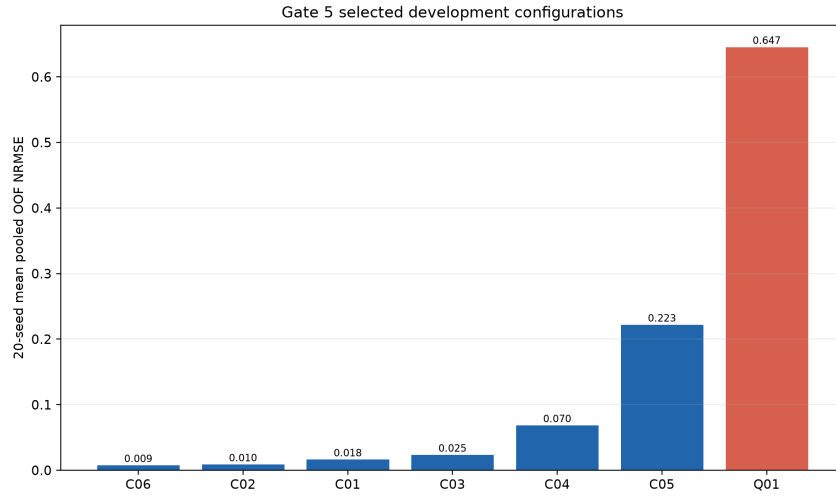


Fig. 4. Twenty-seed summary of selected Gate 5 models (RFIG-022). Seed variation is reported as a stability diagnostic, not as final-test evidence.

The FQK classification track separates discrimination from a safety-filter interpretation. It achieved AUROC 0.7436 and Brier score 0.1561, but recall at the fixed threshold was only 0.1089. Because the positive label is “independently propagated feasible,” this means FQK retained only about 10.9% of actually feasible plans and rejected the remaining 89.1%. That low utility prevents promotion to the proposed screening role. It does not by itself measure unsafe acceptance, which instead requires precision and false-positive analysis. The endpoint plot in Fig. 8 keeps these meanings separate. It is not valid to call FQK useful for safety merely because one metric is above chance or because a quantum feature map was evaluated.

The recall-first classical feasibility analysis, tracked as CSAFE-RF in the repository, provides a separate lesson for a future protocol. A calibrated logistic model reached mean feasible-case recall 0.804253, false-negative rate 0.195747, and Brier score 0.142231. In plain terms, it retained about 80.4% of actually feasible plans and rejected about 19.6% of them. This exposes a tradeoff between calibrated probabilities and retaining useful options, but it does not establish safety: a future screening protocol must separately freeze a feasible-case recall floor, an infeasible-case false-positive limit, and a calibration requirement. CSAFE-RF was not used to re-rank QML candidates, change a threshold, or authorize a mission filter.

5.4 Closed successor branch D034–D049

The closed successor branch contains 16 prospectively bounded candidates. All read the development scope only, used 39,000 development rows per campaign,

No regime cell satisfied every preregistered condition

Preregistered Gate 5 trigger: FAIL. Calibration reads = 0; final-test reads = 0.

Fig. 5. Preregistered algorithm-trigger regime controls (RFIG-023). A passing development trigger would have supported consideration of one further algorithm study, not quantum advantage or operational deployment.

and completed with the recorded status `complete_valid_negative`: the computation and integrity checks succeeded, but the scientific advancement rule did not. The complete results are listed in Table 3. Candidate names in the table are traceability IDs for the frozen protocols, not terminology needed to interpret the conclusion. The table is intentionally included in full rather than selecting only candidates with lower raw error than C06.

Several variants have a lower mean NRMSE than C06. This is a reason to inspect the controls, not a reason to announce a quantum-specific result. The best raw C06 comparison was decision D046’s eight-qubit orthogonalized residual fidelity–RBF kernel (ORFRK-08-R2): NRMSE 0.0064363 versus 0.0068328 for C06, an apparent 5.80% reduction. Both the candidate and its matched classical control first removed residual structure captured by the same RBF stage. The candidate then used a quantum-fidelity channel, while the TWO-RBF control used a second classical RBF channel. TWO-RBF attained 0.0067501, so the candidate’s improvement over that stronger like-for-like control was about 4.65%, below the 5% rule fixed before the run. The result therefore does not meet the evidence standard for a quantum-specific candidate.

5.5 Evidence accounting and closure

Evidence accounting confirms that the reported conclusion follows the authorized boundary. The preregistered development campaign (D006) completed 871 of 871 tasks, with zero task failures and zero reads of calibration or final-test data. The projected-kernel exploratory campaign (D011-R1) read 39,000 development rows

Table 2. Primary and exploratory QML endpoints from the frozen evidence package. All values are development-only. Lower NRMSE and Brier score are better; higher recall is better for retaining actually feasible plans. The feasibility endpoint uses the fixed 0.5 threshold.

Stage	Model and metric	Development result and meaning
Primary Gate 5	Fidelity quantum kernel (Q01); mean NRMSE	Q01: 0.646614; physics-residual C06: 0.008739. Protocol FAIL; no predefined subgroup passed.
Projected follow-up	Projected quantum kernel (Q01b; P001); mean NRMSE	Q01b: 0.661207; C06: 0.006833. Exploratory negative; about 97 times the C06 error.
Feasibility follow-up	Feasibility-only quantum kernel (FK; P001); recall at 0.5	Recall: 0.108860. It retained only 10.9% of actually feasible cases, and classical controls were stronger.
Future lesson	Calibrated logistic model; feasible-case recall and Brier score	Recall: 0.804253; Brier: 0.142231. Protocol-design signal, not a safety result.

Table 3. Closed D034–D049 successor branch. Candidate names are immutable protocol IDs; their definitions are preserved in the repository. Every row is a valid development-only negative. The C06 NRMSE was 0.006833 throughout. Lower NRMSE is better, and a negative paired difference favors the candidate over C06, but each branch also had to pass its matched-control and stopping rules.

Decision	Candidate	Candidate NRMSE	Difference vs C06	95% paired interval
D034	PRQK-08-N	0.029326	0.022493	[0.022486, 0.022500]
D035	CFQSR-08-N	0.013463	0.006630	[0.006627, 0.006632]
D036	TAPQK-08	0.010382	0.003549	[0.003547, 0.003551]
D037	TSQR-08-L025	0.006848	0.000015	[0.000015, 0.000015]
D038	GFRK-08-L010	0.006647	-0.000186	[-0.000187, -0.000184]
D039	EC-GFRK-08-L010	0.006466	-0.000366	[-0.000368, -0.000365]
D040	CE-GFRK-08-L010	0.006504	-0.000329	[-0.000330, -0.000328]
D041	HEFRK-08-E050	0.006592	-0.000241	[-0.000242, -0.000240]
D042	AGEFRK-08-ADAPT	0.006524	-0.000309	[-0.000310, -0.000308]
D043	SFRK-08-CV	0.006445	-0.000387	[-0.000389, -0.000386]
D044	NIFRK-08-NL	0.006751	-0.000081	[-0.000084, -0.000079]
D045	MSFRK-ALL	0.006731	-0.000102	[-0.000103, -0.000100]
D046	ORFRK-08-R2	0.006436	-0.000397	[-0.000399, -0.000394]
D047	OMFRK-ALL	0.006952	0.000119	[0.000113, 0.000125]
D048	NSORFRK-08	0.006590	-0.000243	[-0.000244, -0.000242]
D049	SSORFRK-08	0.006566	-0.000266	[-0.000268, -0.000265]

and zero locked rows. No hardware quantum job, mission-loop run, Gate 6 run, or final-test evaluation occurred.

Figure 11 summarizes the closure recommendation. It reports that no candidate is eligible for Gate 6; it does not represent an unexecuted Gate 6 experiment. Closure decision D055-C then ended active experimentation so that future protocol ideas could not be mistaken for completed evidence during publication preparation.

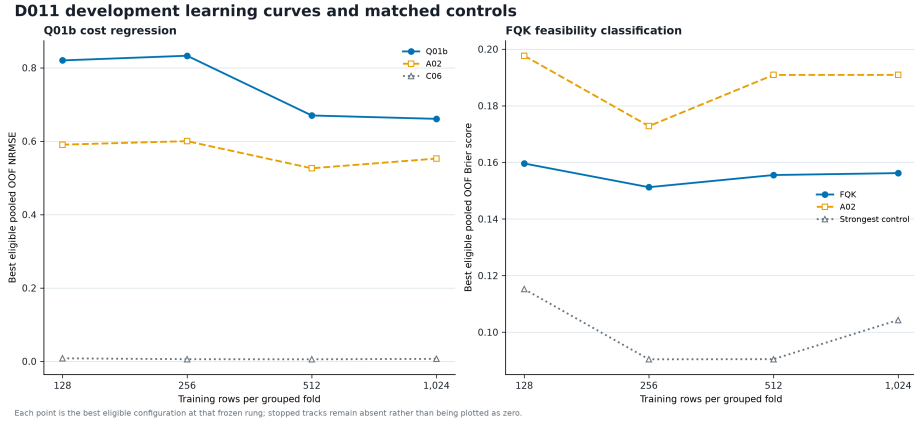


Fig. 6. Projected-kernel follow-up Q01b and matched controls (decision D011 under protocol P001; RFIG-026). The curves show development-only error as training sample size increases. They do not revise the Gate 5 decision.

6 Discussion

The principal finding is a controlled negative benchmark: the runs completed validly, but the tested performance hypotheses were rejected. The Q01 fidelity-style quantum kernel was far behind the physics-residual control, and the Q01b projected feature map did not close the gap. The feasibility-only kernel did not meet its fixed multi-metric screening criteria. The successor branch demonstrates an additional failure mode that matters beyond this task: a candidate can improve over one baseline yet fail against a classical model with the same staged construction. That distinction is why matched dequantization controls and thresholds fixed before outcomes belong in the protocol.

The result is scientifically useful because the negative conclusion is specific. It rules out a family of tested feature maps and successor constructions under a particular public-data scenario generator, grouped development procedure, resource envelope, and comparator set. It does not rule out all QML, all cislunar machine learning, or all future quantum hardware. Such broad claims would require evidence absent here, including separate tasks, new preregistration, scalability studies, noise analysis, hardware measurements, and appropriate classical resource comparisons.

The physics-residual control is a key methodological result. A generic classical baseline is not sufficient when the prediction target contains structure that a low-fidelity model can represent directly. C06’s performance shows that, for this benchmark, the available physical residual structure is more informative than the tested quantum fidelity features. That observation does not establish a theorem about quantum circuits; it establishes a task-specific comparison of inductive bias, the kinds of patterns each model is built to represent, that future proposals must beat fairly.

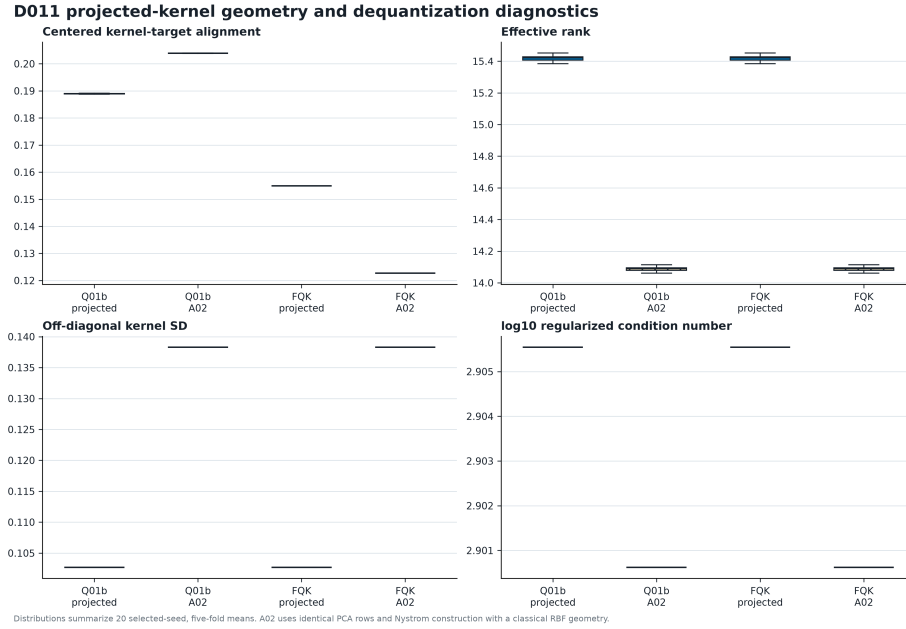


Fig. 7. Projected-kernel geometry and dequantization diagnostics (RFIG-027). Selected quantum-derived similarities are compared with the A02 classical RBF kernel on the same training-fold PCA rows and Nyström landmarks. This tests whether a classical distance kernel explains the apparent geometry.

The retained Q02/Q03 status also matters. Many reports collapse all non-advancing paths into a single “failure” category. Here, the first-rung tasks ran correctly, but the candidates were ineligible for larger-rung computation under a rule frozen before results. Keeping `not_reached_under_frozen_eligibility` separate from a task failure makes the evidence ledger more truthful and prevents a later analyst from inventing a selection path that never existed.

The paper’s societal relevance is therefore not a claim of operational benefit. It is a reusable, human-centered research pattern for high-consequence AI: include an eligible simulator-validation scope, freeze data transformations and groups, use a physics-aware control, compare against matched classical alternatives, retain null outcomes, and make the claim boundary checkable from source artifacts. This helps researchers, reviewers, and future decision-tool designers distinguish a reproducible negative result from evidence for automation. Its value is in reducing the risk that an exciting but unsupported model claim is presented as a safe decision-support capability.

The HCI implication is deliberately limited. The present results do not show that a planner would understand, trust, reject, or use the evidence package well. They specify what a later human-centered evaluation should not omit:

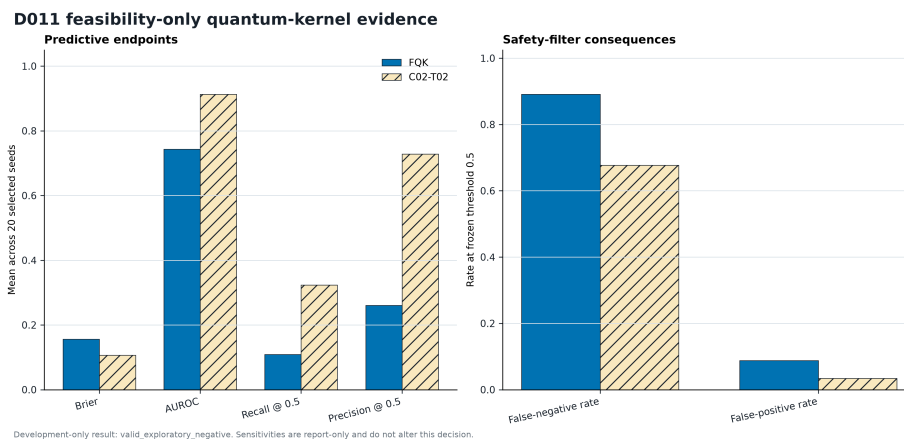


Fig. 8. Feasibility-only quantum-kernel (FQK) endpoints (RFIG-028). Brier score, AUROC, feasible-case recall, precision, and consequence metrics are compared with the strongest complete control. The fixed screening criteria did not pass.

provenance, model scope, uncertainty, failure visibility, and a non-automated fallback. Testing those requirements with intended users remains future work.

6.1 Implications for QML benchmarking

The QML literature has repeatedly shown that benchmark conclusions depend on which classical methods are made available to the comparison [1,15]. The D046 result gives a concrete trajectory application example. Comparing only with C06 would make the candidate’s raw 5.80% difference look promising. The matched TWO-RBF alternative explains enough of that difference that the candidate fails the frozen 5% rule. The interpretation changes because the question changes from “is the candidate better than one baseline?” to “is its incremental mechanism still supported when a classical model is given the same residual construction?”

This contrast does not show that the matched control is the universally best classical model. It shows why a QML result needs multiple controls chosen to test the claimed mechanism. A fair comparison is not one that intentionally makes every model identical; it is one that removes alternative explanations for the purported QML contribution. In this benchmark, the alternative explanation was a classical residual representation, and it remained competitive.

6.2 Safety objective as a distinct scientific question

The FQK and CSAFE-RF records demonstrate that predictive metrics must be tied to both label meaning and intended action. In these executed analyses, the positive label is feasible. AUROC measures overall ranking, Brier score measures probability error, and recall measures how many actually feasible options

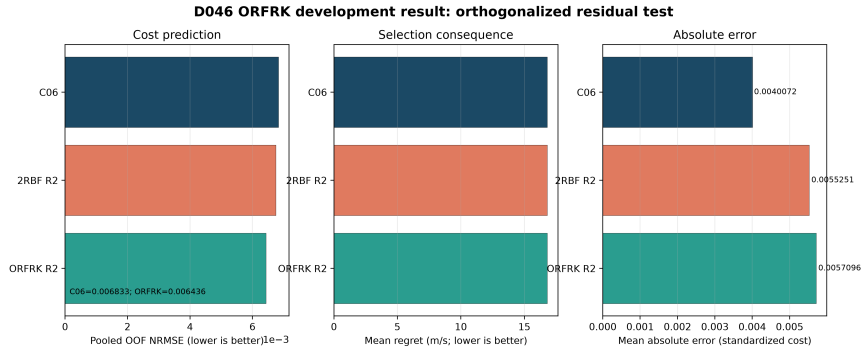


Fig. 9. D046 ORFRK result comparison (RFIG-078). Here $q = 8$ means an eight-qubit statevector simulation. The candidate and classical control share the first RBF correction and differ only in the second-stage channel. Lower raw error than C06 is therefore not sufficient for a QML-specific claim.

are retained. Precision and the false-positive rate address the different risk of admitting an actually infeasible option. FQK remains negative under its frozen multi-metric rule even though its AUROC is above random, because its Brier score and feasible-case recall did not match the classical controls. This guards against promoting a model by whichever metric looks most favorable afterward.

The higher-recall classical result is not a rescue or a safety result. It is a future design lesson: a screening protocol should define, before fitting, a minimum feasible-case recall for usefulness, a maximum infeasible-case false-positive rate for protection, a calibration requirement, and the action taken when the model abstains. Those quantities cannot be replaced by a single “safety” score. Testing such a policy requires a separately authorized experiment; using it to alter Gate 5 after inspection would invalidate the original test.

7 Limitations and Threats to Validity

First, the study is public-data and development-only. Calibration and final-test data were not read. The paper cannot estimate final-test error, operational calibration, mission-loop behavior, propellant savings, or flight readiness. The use of Artemis-related public work establishes relevance of the problem framing, not access to mission-owned truth data or authority to draw mission conclusions.

Second, simulator agreement is bounded. The Gate 3 results support the eligible public endpoint comparisons, but they do not validate every force model, contingency, navigation product, or human-rating consideration. RTC3 is source-ineligible, and some generated scenario decisions do not have an independently feasible numerical reference. Those limitations remain in the dataset rather than being removed after model results are observed.

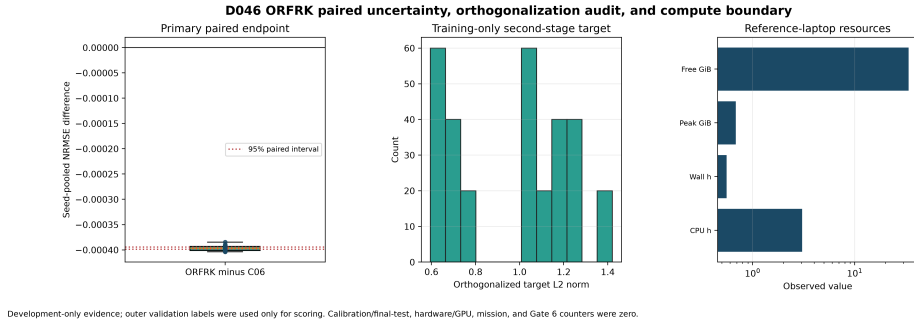


Fig. 10. D046 paired audit and resource boundary (RFIG-079). The figure reports matched-seed error differences, checks that residual removal used training data only, data-integrity evidence, and reference-laptop resources. It contains no quantum-hardware or GPU result.

Third, every quantum experiment is an exact classical statevector simulation with at most eight qubits, run within a laptop-bounded resource envelope. The runs therefore omit finite measurement sampling (shot noise), imperfect gates, device-connection limits, the time required to prepare an encoded state, and transpilation, which maps an abstract circuit to a particular processor. They report neither hardware calibration nor end-to-end quantum runtime and cannot demonstrate quantum speedup, hardware efficiency, or practical quantum advantage. Any such claim would require these costs to be measured against a fair classical resource baseline.

Fourth, the methods are limited by their frozen design. The chosen preprocessing, feature maps, sample rungs, controls, and thresholds make the comparison interpretable but restrict the scope of inference. A different method can be scientifically tested only through a new protocol, not by modifying the completed campaign until a favorable result appears.

Finally, the literature synthesis is bounded, not systematic. The inclusion set provides methodological context and control requirements but does not prove that no related method exists. This is why the paper makes a benchmark claim rather than a novelty or universal-performance claim.

8 Future Research Boundary

The negative results motivate questions about task-specific symmetry, representation, and residual structure, but the completed evidence does not answer them. A future geometric-QML direction was discussed as a hypothesis, not as a result. Here a symmetry would be a transformation, such as rotating all relevant vectors together, that preserves the physical relationship between inputs and labels. An equivariant model is designed so that its output changes consistently under such a transformation. The first unanswered question is whether any non-trivial symmetry actually survives fixed Earth–Moon–Sun ephemerides, thrust

D025-C Gate 5 closure: no QML candidate for Gate 6

All Gate 5 and post-Gate-5 evidence is development-only or reporting-only; Gate 6 remains locked.

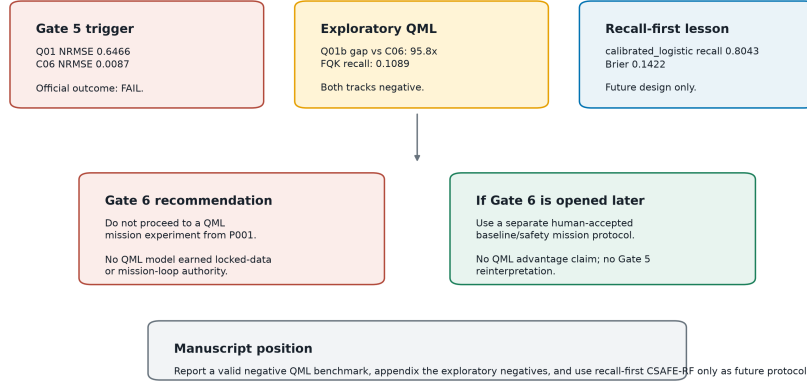


Fig. 11. Gate 5 closure and Gate 6 recommendation (RFIG-044). No QML candidate from the closed evidence branches is eligible for Gate 6. This is a closure recommendation, not a Gate 6 result.

constraints, and reference-frame choices. Equivariance alone would not imply a QML advantage over classical equivariant models.

A future protocol would require a mathematical contract, a symmetry audit, baseline reproduction, a representation- and parameter-matched classical comparison, ablations of symmetry and initialization, resource accounting, and explicit rejection criteria. The D054-A/P018 binding freeze records how a future audit could specify its representation, dynamical Lie algebra (DLA) basis, allowed circuit generators, development-only case groups, numerical tolerance, output paths, and figures. A DLA describes the transformations that can be generated by the allowed circuit operations. Closure decision D055-C declined to execute the audit. Accordingly, this manuscript contains no DOP853 trajectory replay (a high-order numerical integration check), no singular-value-decomposition (SVD) rank or nullspace result, no new symmetry dataset, no variational circuit training, and no geometric-QML performance claim.

This boundary is not a weakness in reporting. It prevents a promising idea from being laundered into evidence through narrative. The appropriate next step, if future authors choose to reopen research, is a separately approved protocol that can reject the hypothesis as readily as it can support it.

9 Data, Code, and Figure Availability

The source repository, governance records, machine-readable reporting outputs, figure registry, and the published source release are available at the project repository. The published release record is release v0.3.0. This paper package

copies the exact figures used in the manuscript and the three core result tables plus figure registry. The package does not include calibration or final-test rows, mission-owned evidence, trained weights, or unexecuted future-protocol outputs.

10 Conclusion

Under the frozen public-data development benchmark, the tested QML candidates did not outperform strong classical controls, and no tested candidate is eligible for a Gate 6 mission experiment. The primary whole-state fidelity kernel (Q01), projected-kernel follow-up (Q01b), feasibility-only kernel (FQK), and all 16 bounded successors produced negative outcomes under their stated criteria. The strongest successor appeared better than the C06 physics-residual control, but its advantage over a like-for-like classical alternative was only 4.65%, below the 5% threshold fixed before execution.

The study’s contribution is a reproducible negative result with an explicit governance and human-centered claim structure. It reports what was tested, what did not advance, what remains a future lesson, and what a human reviewer would need to see before considering further authorization. Its practical societal value is a more accountable way to evaluate AI evidence for high-consequence decision contexts, not a deployed mission capability. It does not establish quantum advantage, fuel savings, mission utility, hardware performance, flight readiness, NASA approval, or an invented QML model. That restraint is necessary for the result to be useful.

Acknowledgments. Reserved for an advisor contribution statement, pending advisor review.

Appendix: Reproducibility and Evidence Ledger

The paper package is structured so that a reviewer can trace every main-text number to a copied source table or figure. The Gate 5 CSV and the successor branch-summary CSV (comma-separated-value) files in `results_tables/` provide the numerical values reported in Tables 2 and 3. The copied figure registry records the title, evidence status, source data, generator, digest, and claim boundary for every figure included here. The figures retain their registry identifiers in their captions.

The release package provides scripts, fixed configuration records, data cards, and decision logs. Reproduction of the published development figures should begin from the tagged repository state and should preserve the lock on calibration and final-test data. Re-running a report-generation script is different from authorizing new fitting or a new experimental campaign.

Interpretation Checklist for Reviewers

The following checklist summarizes the intended reading of the paper:

Table 4. Literature-source audit. The table distinguishes protocol influence from direct empirical evidence in this paper. All cited records constrain interpretation; none supplies a result for the present benchmark.

Source category	Role in the experiment and manuscript
Human-rated mission and credibility	Target constraints, intended-use boundary, simulator credibility, and separate crew requirements: [22,8,9].
Classical trajectory and feasibility learning	Classical surrogate, reachability, low-fidelity reference, and nonconvergence precedents: [24,20,7,19,23,25].
QML family and trainability	Feature-space framing, data re-uploading, quantum-agent controls, encoding, and initialization: [16,5,11,18,2,4].
Kernel comparison and dequantization	Kernel design, bandwidth, data dependence, random features, and dequantization controls: [10,17,6,14,21].
Benchmark validity and reporting	Fair comparison, scrutiny, seed/generalization discipline, and error-distribution reporting: [1,15,13,12].
Adjacent quantum optimization	Quantum annealing for trajectory transcription was reviewed but deliberately kept outside the QML benchmark: [3].

1. Treat Gate 3 as an eligible-scope simulator consistency result, not a mission validation result.
2. Treat Gate 5 as rejection of the Q01 fidelity-kernel hypothesis on this development benchmark, not a universal statement that QML cannot work.
3. Treat the projected Q01b kernel, feasibility-only FQK classifier, and 16 successors D034–D049 as development-only negative evidence that cannot revise Gate 5 or open Gate 6.
4. Treat D046 as evidence that an apparent improvement over the C06 physics-residual model must also be tested against a like-for-like classical control.
5. Treat the recall-first analysis as a future lesson about retaining feasible plans while controlling infeasible admissions, not as a deployed safety filter.
6. Do not infer quantum advantage, hardware performance, propellant saving, or flight readiness from any figure or table in this package.

Literature-Source Audit

The literature review was re-audited against the completed experiment. Table 4 accounts for the 23 accepted records in the repository extraction matrix, plus the random-Fourier source that underpins the implemented A01 control and the quantum-kernel benchmark source retained for the P001 discussion. The full extraction matrix remains the provenance record; the bibliography contains only sources that support an explicit statement, design choice, control, or scope decision in this manuscript.

The audit does not convert prior literature into evidence for the present models. In particular, variational QML, quantum reinforcement learning, and quantum annealing sources are cited to explain why a model family was frozen, controlled, or excluded. The empirical findings remain those in the registered public-data development campaigns described in this paper.

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